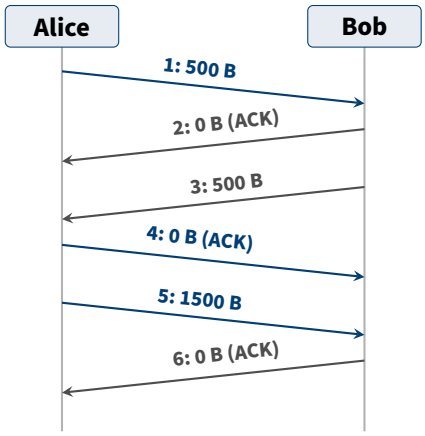
Aufgabe 1: Übertragung in TCP

TCP-Übertragung & Sequenznummern – Übungsblatt

Bestimmen Sie für die folgenden TCP-Segmentabfolgen jeweils die Sequenz- (`Seq`) und Bestätigungsnummern (`Ack`) der beteiligten Hosts Alice und Bob.
Regeln: `Seq` entspricht dem Byte-Offset der Nutzdaten im TCP-Datenstrom. `Ack` bestätigt den lückenlosen Empfang aller Bytes bis zu dieser Nummer (Ack $=$ Seq $+$ Nutzdaten). Reine Bestätigungspakete (0 Byte Nutzdaten) verbrauchen keine Sequenznummern.

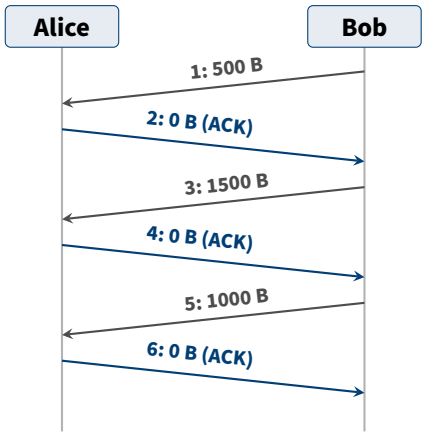
1.) Initial Sequence Numbers: Alice ISN = `601`, Bob ISN = `1001`



Nr.	Richtung	Nutzdaten	Seq	Ack
1	Alice → Bob	500 B	601	1001
2	Bob → Alice	0 B (ACK)	_____	_____
3	Bob → Alice	500 B	_____	_____
4	Alice → Bob	0 B (ACK)	_____	_____
5	Alice → Bob	1500 B	_____	_____
6	Bob → Alice	0 B (ACK)	_____	_____

TCP Simulator ↗
Interaktive Ablaufverfolgung

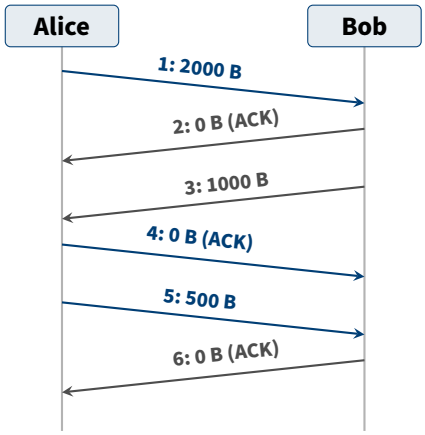
2.) **Initial Sequence Numbers:** Alice ISN = `201`, Bob ISN = `3501`



Nr.	Richtung	Nutzdaten	Seq	Ack
1	Bob → Alice	500 B	3501	201
2	Alice → Bob	0 B (ACK)	_____	_____
3	Bob → Alice	1500 B	_____	_____
4	Alice → Bob	0 B (ACK)	_____	_____
5	Bob → Alice	1000 B	_____	_____
6	Alice → Bob	0 B (ACK)	_____	_____

TCP Simulator ↗
Interaktive Ablaufverfolgung

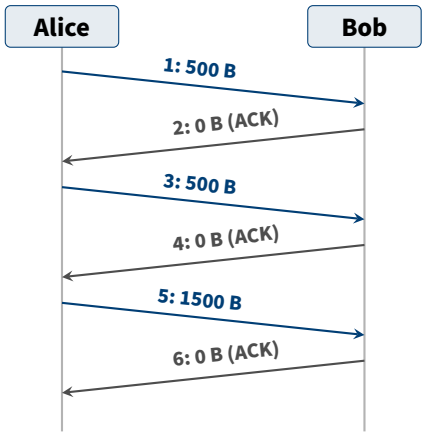
3.) **Initial Sequence Numbers:** Alice ISN = `501`, Bob ISN = `501`



Nr.	Richtung	Nutzdaten	Seq	Ack
1	Alice → Bob	2000 B	501	501
2	Bob → Alice	0 B (ACK)	_____	_____
3	Bob → Alice	1000 B	_____	_____
4	Alice → Bob	0 B (ACK)	_____	_____
5	Alice → Bob	500 B	_____	_____
6	Bob → Alice	0 B (ACK)	_____	_____

TCP Simulator ↗
Interaktive Ablaufverfolgung

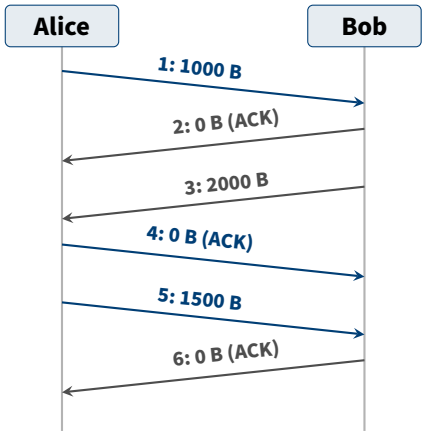
4.) **Initial Sequence Numbers:** Alice ISN = `701`, Bob ISN = `2501`



Nr.	Richtung	Nutzdaten	Seq	Ack
1	Alice → Bob	500 B	701	2501
2	Bob → Alice	0 B (ACK)	_____	_____
3	Alice → Bob	500 B	_____	_____
4	Bob → Alice	0 B (ACK)	_____	_____
5	Alice → Bob	1500 B	_____	_____
6	Bob → Alice	0 B (ACK)	_____	_____

TCP Simulator ↗
Interaktive Ablaufverfolgung

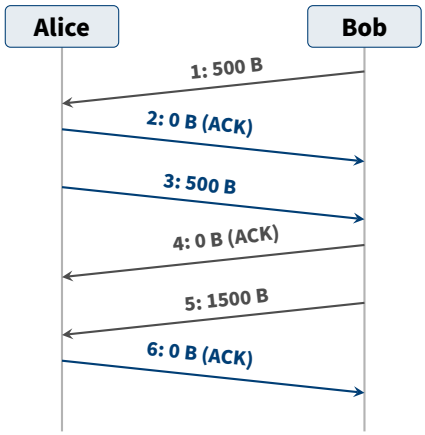
5.) **Initial Sequence Numbers:** Alice ISN = `201`, Bob ISN = `501`



Nr.	Richtung	Nutzdaten	Seq	Ack
1	Alice → Bob	1000 B	201	501
2	Bob → Alice	0 B (ACK)	_____	_____
3	Bob → Alice	2000 B	_____	_____
4	Alice → Bob	0 B (ACK)	_____	_____
5	Alice → Bob	1500 B	_____	_____
6	Bob → Alice	0 B (ACK)	_____	_____

TCP Simulator ↗
Interaktive Ablaufverfolgung

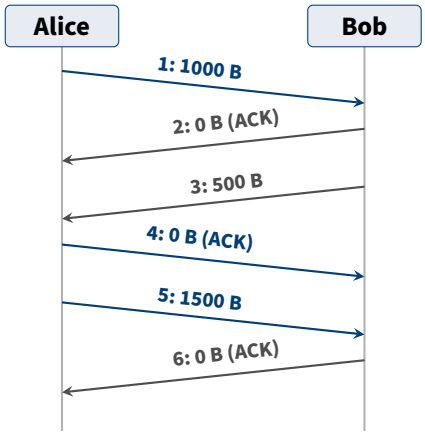
6.) **Initial Sequence Numbers:** Alice ISN = `601`, Bob ISN = `2001`



Nr.	Richtung	Nutzdaten	Seq	Ack
1	Bob → Alice	500 B	2001	601
2	Alice → Bob	0 B (ACK)	_____	_____
3	Alice → Bob	500 B	_____	_____
4	Bob → Alice	0 B (ACK)	_____	_____
5	Bob → Alice	1500 B	_____	_____
6	Alice → Bob	0 B (ACK)	_____	_____

TCP Simulator ↗
Interaktive Ablaufverfolgung

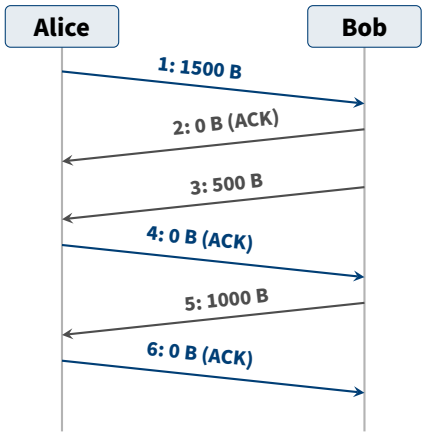
7.) **Initial Sequence Numbers:** Alice ISN = `201`, Bob ISN = `3501`



Nr.	Richtung	Nutzdaten	Seq	Ack
1	Alice → Bob	1000 B	201	3501
2	Bob → Alice	0 B (ACK)	_____	_____
3	Bob → Alice	500 B	_____	_____
4	Alice → Bob	0 B (ACK)	_____	_____
5	Alice → Bob	1500 B	_____	_____
6	Bob → Alice	0 B (ACK)	_____	_____

TCP Simulator ↗
Interaktive Ablaufverfolgung

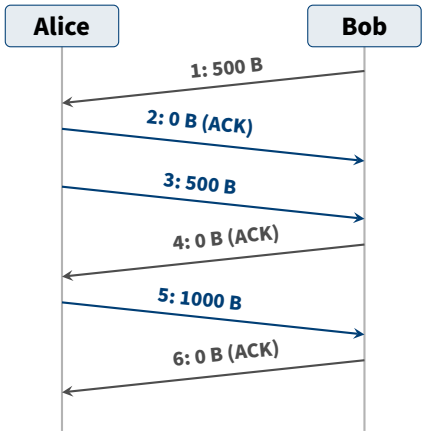
8.) Initial Sequence Numbers: Alice ISN = `101`, Bob ISN = `2001`



Nr.	Richtung	Nutzdaten	Seq	Ack
1	Alice → Bob	1500 B	101	2001
2	Bob → Alice	0 B (ACK)	_____	_____
3	Bob → Alice	500 B	_____	_____
4	Alice → Bob	0 B (ACK)	_____	_____
5	Bob → Alice	1000 B	_____	_____
6	Alice → Bob	0 B (ACK)	_____	_____

TCP Simulator ↗
Interaktive Ablaufverfolgung

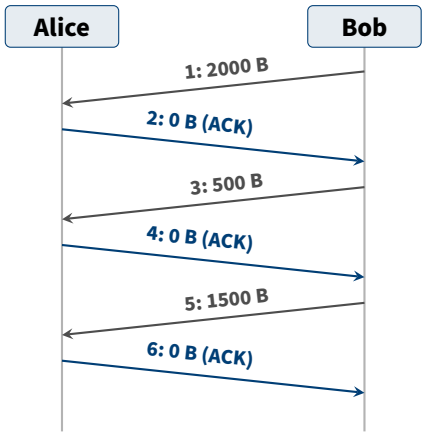
9.) Initial Sequence Numbers: Alice ISN = `101`, Bob ISN = `3001`



Nr.	Richtung	Nutzdaten	Seq	Ack
1	Bob → Alice	500 B	3001	101
2	Alice → Bob	0 B (ACK)	_____	_____
3	Alice → Bob	500 B	_____	_____
4	Bob → Alice	0 B (ACK)	_____	_____
5	Alice → Bob	1000 B	_____	_____
6	Bob → Alice	0 B (ACK)	_____	_____

TCP Simulator ↗
Interaktive Ablaufverfolgung

10.) Initial Sequence Numbers: Alice ISN = `401`, Bob ISN = `501`



Nr.	Richtung	Nutzdaten	Seq	Ack
1	Bob → Alice	2000 B	501	401
2	Alice → Bob	0 B (ACK)	_____	_____
3	Bob → Alice	500 B	_____	_____
4	Alice → Bob	0 B (ACK)	_____	_____
5	Bob → Alice	1500 B	_____	_____
6	Alice → Bob	0 B (ACK)	_____	_____

TCP Simulator ↗

Interaktive Ablaufverfolgung